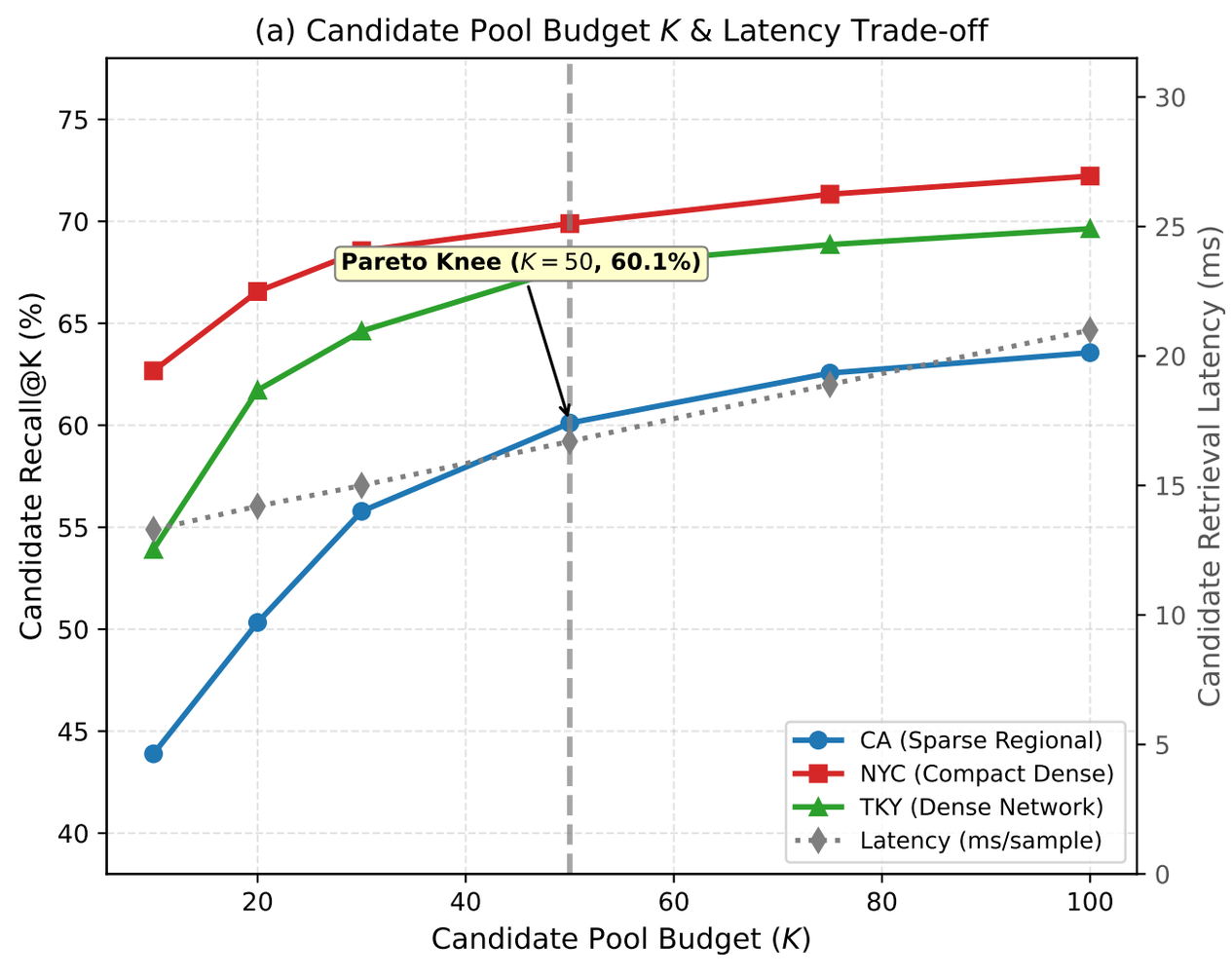
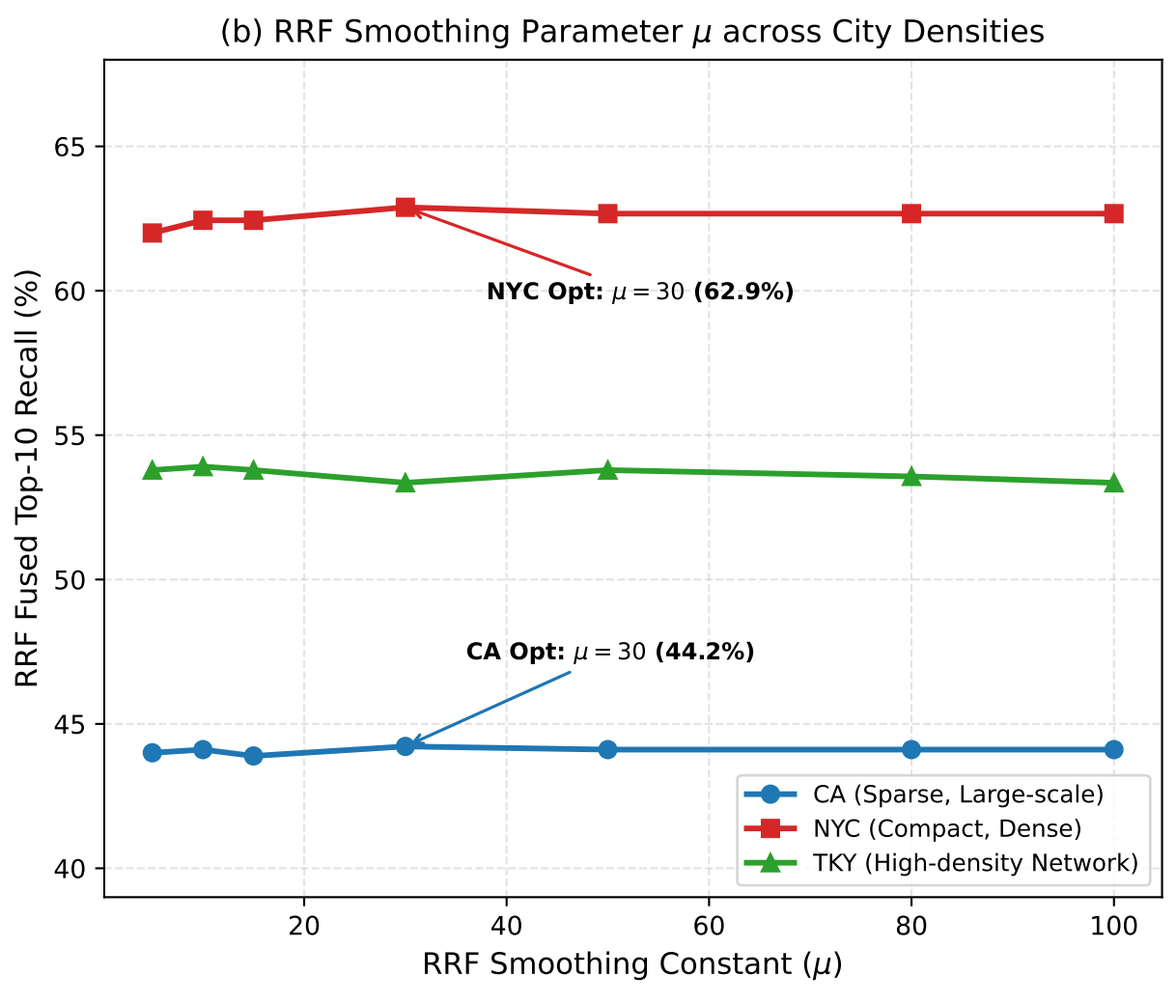
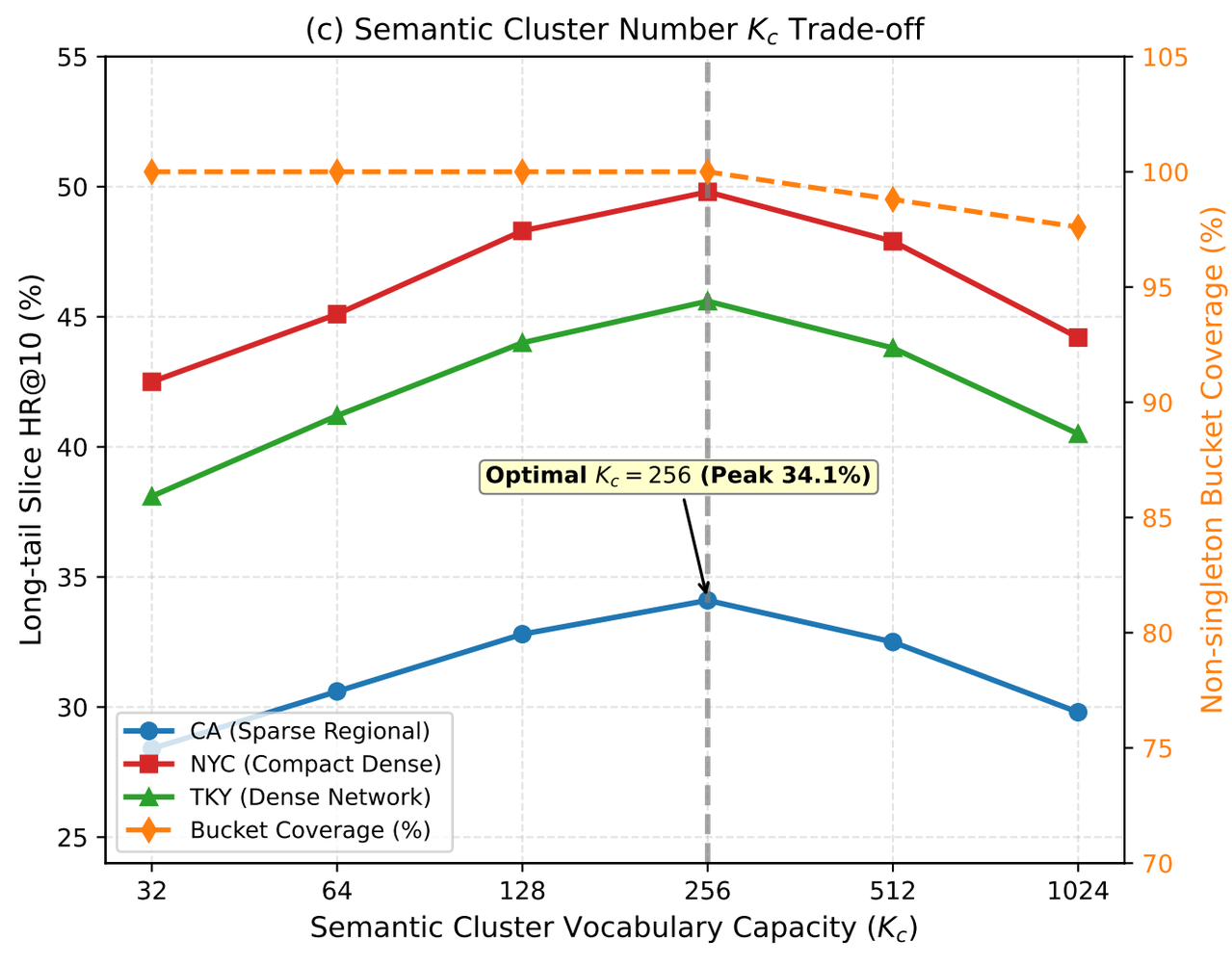
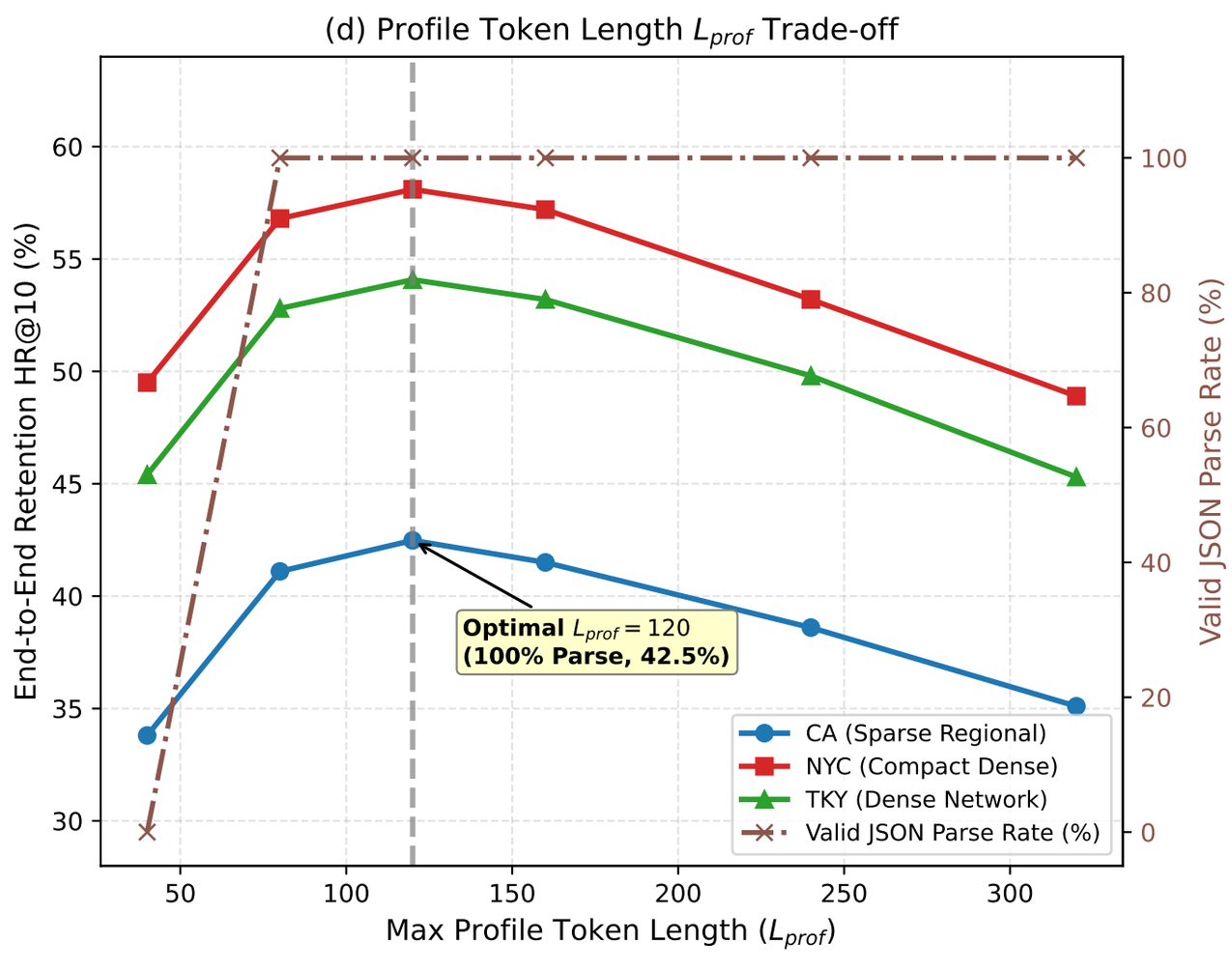

(a) Candidate Pool Budget K & Latency Trade-off
Pareto Knee ($K = 50$, 60.1%)
Candidate Recall@K (%)
Candidate Retrieval Latency (ms)
Candidate Pool Budget (K)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Latency (ms/sample)


(b) RRF Smoothing Parameter μ across City Densities
NYC Opt: $\mu = 30$ (62.9%)
CA Opt: $\mu = 30$ (44.2%)
RRF Fused Top-10 Recall (%)
RRF Smoothing Constant (μ)
CA (Sparse, Large-scale)
NYC (Compact, Dense)
TKY (High-density Network)


(c) Semantic Cluster Number K_c Trade-off
Optimal $K_c = 256$ (Peak 34.1%)
Long-tail Slice HR@10 (%)
Non-singleton Bucket Coverage (%)
Semantic Cluster Vocabulary Capacity (K_c)
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Bucket Coverage (%)


(d) Profile Token Length L_{prof} Trade-off
Optimal $L_{prof} = 120$
(100% Parse, 42.5%)
End-to-End Retention HR@10 (%)
Valid JSON Parse Rate (%)
Max Profile Token Length (L_{prof})
CA (Sparse Regional)
NYC (Compact Dense)
TKY (Dense Network)
Valid JSON Parse Rate (%)